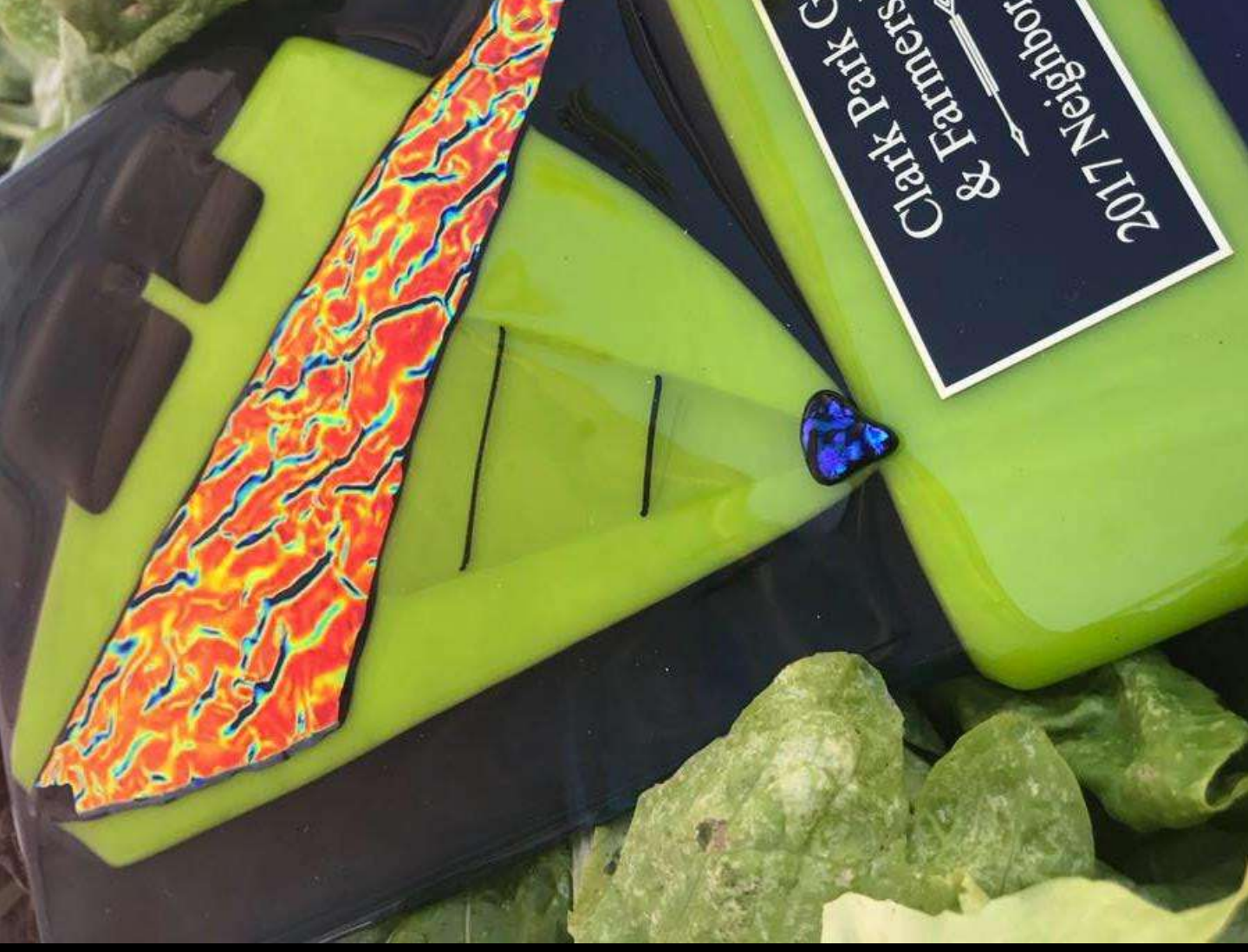

DESIGN REVIEW

STAGE 1: CLARK PARK COMPOSTING





TEAM MEMBERS

Nathan Amyot - Team Lead, Biomedical Engineering

Cohen Jeffries - Meeting Organizer, Biomedical Engineering

Heather Taylor - Documentation Lead, Biomedical Engineering

Patrick "Riley" O'Hara - Prototyping Lead, Mechanical Engineering

Emma Tapia - Communication Lead, Mechanical Engineering

Genevieve Alexander - Research Lead, Environmental Engineering

James "Colton" Davis - Design Lead, Biomedical Engineering

Denise Hill - Budget Lead, Mechanical Engineering

Casey Boyd – Community Partner



BACKGROUND

- Tempe Arizona, founded 20
 - Volunteer-run, non-profit ga
 - Community Bonds
 - Sustainable Gardening
 - Educational Opportunities
 - Local Artisan Markets
-

WHAT'S THE PROBLEM?

Currently there is **no functioning compost system** at Clark Park Gardens, which does not align with their goals of being environmentally friendly and ends up costing them money to have their plant waste removed by the city of Tempe.

STAKEHOLDERS

- Volunteers & Gardeners
- Animals/Plants/Environment
- Neighborhood Residents
- School Children



USER NEEDS

1. Easily rotates
2. Produces nutrient rich compost
3. Isolates compost from surrounding environment
4. Does not produce a bad smell
5. Includes an educational aspect



DESIGN REQUIREMENTS

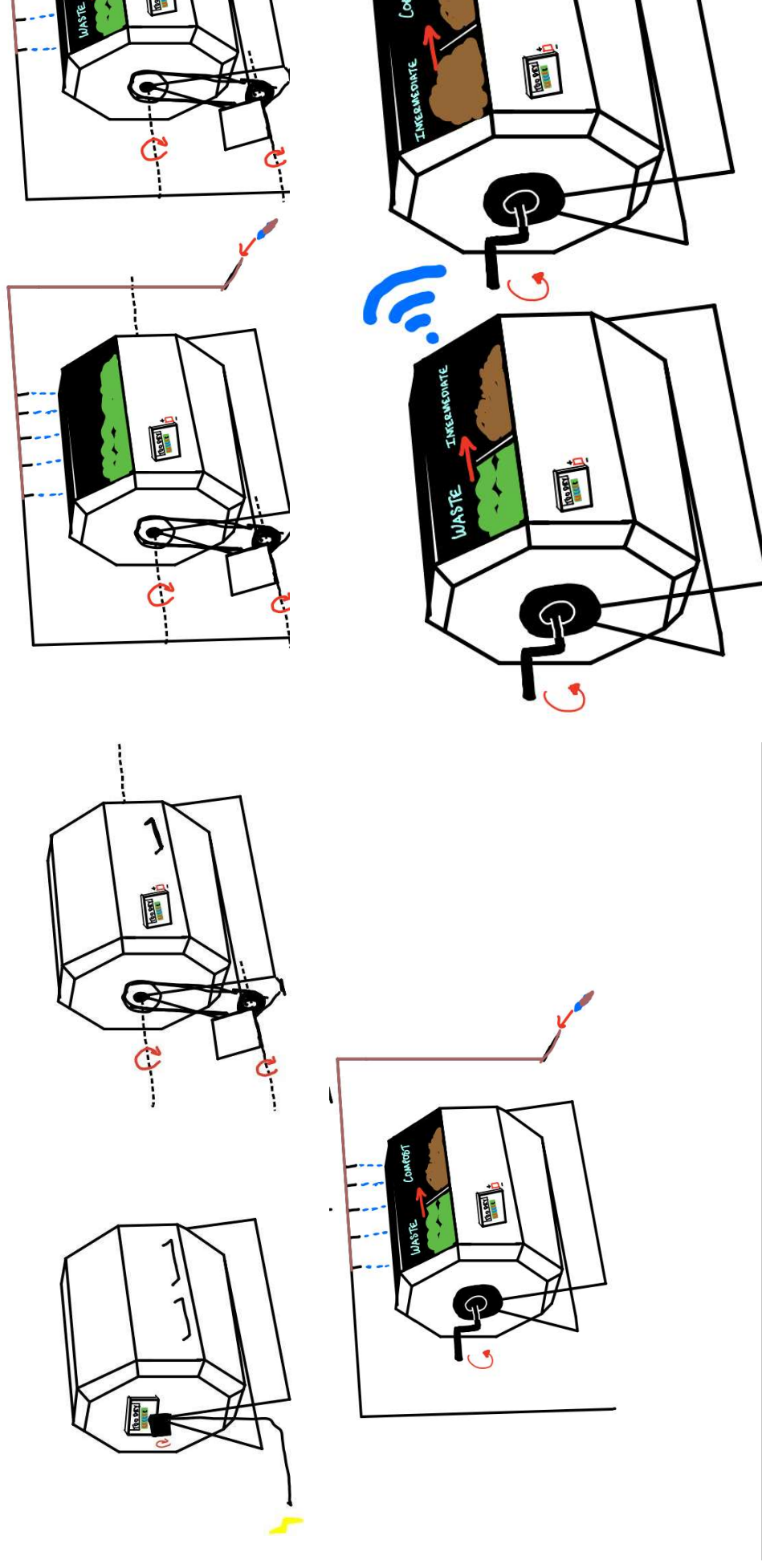


- Need enough usable compost to be produced
 - Need to be able to easily manage the compost
 - Mitigates smell to ensure comfort of grower
 - Need essential nutrients to grow happily
 - Monitors and displays properties of compost
-

COMPETITOR ANALYSIS

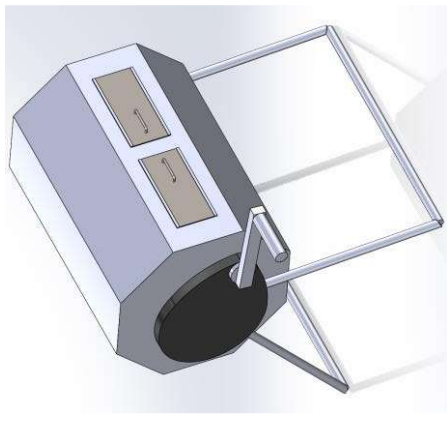
	Size (gal)	Price	Mixing & Ventilation System	Door/Security System
Miracle-Gro Dual Chamber Compost Tumbler	18.5 gal	\$139.90	Air vents on the side, rotating barrel	Sliding doors, lock
Jora Compost Bin Tumbler	70 gal	\$569.00	Air vents on the side, rotating barrel	Sliding doors which lock
Vivosun Outdoor Tumbling Composter	43 gal	\$79.99	Air vents on the side, rotating barrel with fins inside	Sliding doors, lock
Squeeze Master Large Compost Tumbler	43 gal	\$79.99	Air vents on the side, rotating barrel	Small sliding doors, no lock
BLACK+DECKER 40-Gallon Dual Chamber Compost Tumbler Outdoor	40 gal	\$129.99	Air vents on the side, internal mixing bars, crank handle for rotation	Sliding doors, lock

BRAINSTORMING + TOP DESIGNS/SOLUTION



FINAL DESIGN

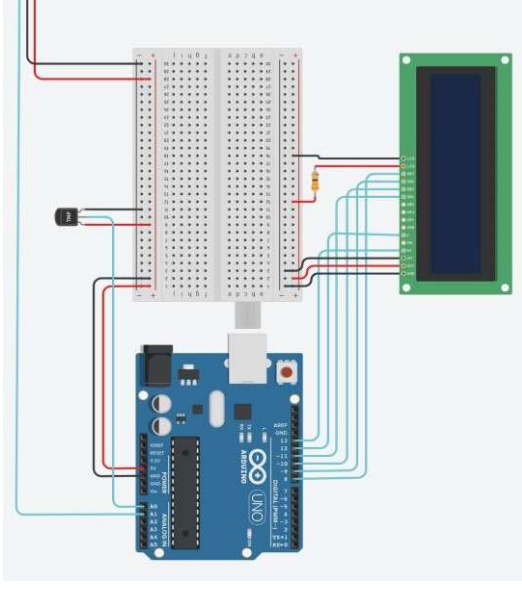
- Using compost tumbler as base
- Plan to integrate Arduino measurement system using ESP32 Board
- Designed hand crank system to improve usability of tumbler



PROTOTYPING PLAN

Current Semester – Spring 2025

- Goal of delivering a working prototype by end of the semester
- Decided to focus on integrating designed measurement system
- Arduino code and circuit designed; materials ready to start building prototype



PROTOTYPING TIMELINE

January-February	March	April	Summer
• Research Project • Meet with Community Partner • Brainstorm Solutions • Develop Arduino circuit and code • Purchase needed materials for prototype	• Second meeting with Community Partner • Design composter • Order materials	• Build prototype • Deliver prototype to Clark Park Garden	• Have garden test composter and feedback next spring

IMPACT OF SOLUTION

Garden
Community's
Compost

Educational
Outreach

Reduction of
City Compost
Service

BUDGET

Item	Quantity	Unit Cost
43 Gallon Compost Bin Outdoor Tumbler	2	\$64.98
MAKERELE ABS Plastic Small Outdoor Waterproof Box 5.9×3.9×2.8 in	2	\$16.20
ESP32 Board	3	\$5.30
Slip Rings	2	\$19.92
Caulk	1	\$13.98
Caulk Gun	1	\$4.78
Total (including shipping & tax)	-	\$268.79

FUTURE STEPS

- Current: Budget approved, materials ordered and arrived
 - Future:
 - Initial prototype finished by March 24th, 2025
 - Design testing by March 31st, 2025
 - Design update / Meet with community partner
 - Finalize design and construct final product by April 14th, 2025
 - Implement design and gather customer feedback by April 28, 2025
-

ENGINEERING + DESIGN CHALLENGES/BARRIERS

- Weatherproofing our ESP 32 controller and attachments while maintaining easy access to the screen.
 - Transporting the final solution from ASU campus to Clark Park Community Garden.
 - Cooling the compost bin during hot Arizona Summers.
 - Automated water delivery to the compost.
 - Educating the park users/volunteers on proper composting strategies.
 - Communicating via the ESP 32 board to the stakeholders phone or email.
-

THANK YOU!

ANY QUESTIONS?
